

DMA022A
Mathematics for AI

Department of Mathematics
JECRC University Jaipur

Unit-I

Matrices - Introduction

Definition:

A matrix is a set or group of numbers arranged in a square or rectangular array enclosed by two brackets

Example:1

$$\begin{bmatrix} 1 & -1 \end{bmatrix}$$

Example:2

$$\begin{bmatrix} 4 & 2 \\ -3 & 0 \end{bmatrix}$$

Example:3

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Matrices - Introduction

Properties:

- A specified number of rows and a specified number of columns
- Two numbers (rows x columns) describe the dimensions or size of the matrix.

Examples:

3x3 matrix	$\begin{bmatrix} 1 & 2 & 4 \\ 4 & -1 & 5 \\ 3 & 3 & 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 3 & -3 \\ 0 & 0 & 3 & 2 \end{bmatrix}$	$\begin{bmatrix} 1 & -1 \end{bmatrix}$
2x4 matrix			
1x2 matrix			

Matrices - Introduction

A matrix is denoted by a bold capital letter and the elements within the matrix are denoted by lower case letters

e.g. matrix $[A]$ with elements a_{ij}

$$A_{m \times n} = \begin{bmatrix} a_{11} & a_{12} \dots & a_{ij} & a_{in} \\ a_{21} & a_{22} \dots & a_{ij} & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{ij} & a_{mn} \end{bmatrix}$$

i goes from 1 to m

j goes from 1 to n

Matrices - Introduction

TYPES OF MATRICES

1. Column matrix or vector:

The number of rows may be any integer but the number of columns is always 1

$$\begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}$$

Matrices - Introduction

2. Row matrix or vector

Any number of columns but only one row

$$\begin{bmatrix} 1 & 1 & 6 \end{bmatrix} \quad \begin{bmatrix} 0 & 3 & 5 & 2 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \end{bmatrix}$$

Matrices - Introduction

3. Rectangular matrix

Contains more than one element and number of rows is not equal to the number of columns

$$\begin{bmatrix} 1 & 1 \\ 3 & 7 \\ 7 & -7 \\ 7 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 0 & 3 & 3 & 0 \end{bmatrix}$$

$$m \neq n$$

Matrices - Introduction

4. Square matrix

The number of rows is equal to the number of columns

(a square matrix $\mathbf{A}$ has an order of m)

$$\begin{bmatrix} 1 & 1 \\ 3 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 \\ 9 & 9 & 0 \\ 6 & 6 & 1 \end{bmatrix} \quad m \times m$$

The principal or main diagonal of a square matrix is composed of all elements a_{ij} for which $i=j$

Matrices - Introduction

5. Diagonal matrix

A square matrix where all the elements are zero except those on the main diagonal

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 9 \end{bmatrix}$$

i.e. $a_{ij} = 0$ for all $i \neq j$

$a_{ij} \neq 0$ for some or all $i = j$

Matrices - Introduction

6. Unit or Identity matrix - I

A diagonal matrix with ones on the main diagonal

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a_{ij} & 0 \\ 0 & a_{ij} \end{bmatrix}$$

i.e. $a_{ij} = 0$ for all $i \neq j$

$a_{ij} = 1$ for some or all $i = j$

Matrices - Introduction

7. Null (zero) matrix - 0

All elements in the matrix are zero

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$a_{ij} = 0 \quad \text{For all } i, j$$

Matrices - Introduction

8. Triangular matrix

A square matrix whose elements above or below the main diagonal are all zero

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 8 & 9 \\ 0 & 1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

Matrices - Introduction

8a. Upper triangular matrix

A square matrix whose elements below the main diagonal are all zero.

$$\begin{bmatrix} a_{ij} & a_{ij} & a_{ij} \\ 0 & a_{ij} & a_{ij} \\ 0 & 0 & a_{ij} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 8 & 7 \\ 0 & 1 & 8 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 7 & 4 & 4 \\ 0 & 1 & 7 & 4 \\ 0 & 0 & 7 & 8 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

i.e. $a_{ij} = 0$ for all $i > j$

Matrices - Introduction

8b. Lower triangular matrix

A square matrix whose elements above the main diagonal are all zero

$$\begin{bmatrix} a_{ij} & 0 & 0 \\ a_{ij} & a_{ij} & 0 \\ a_{ij} & a_{ij} & a_{ij} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 2 & 3 \end{bmatrix}$$

i.e. $a_{ij} = 0$ for all $i < j$

Matrices – Introduction

9. Scalar matrix

A diagonal matrix whose main diagonal elements are equal to the same scalar

A scalar is defined as a single number or constant

$$\begin{bmatrix} a_{ij} & 0 & 0 \\ 0 & a_{ij} & 0 \\ 0 & 0 & a_{ij} \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

i.e. $a_{ij} = 0$ for all $i \neq j$
 $a_{ij} = a$ for all $i = j$

SYMMETRIC MATRICES

A Square matrix is symmetric if it is equal to its transpose:

$$\mathbf{A} = \mathbf{A}^T$$

$$A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$$

$$A^T = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$$

Skew-SYMMETRIC MATRICES

A Square matrix is skew-symmetric if it is equal to its negative transpose:

$$\mathbf{A} = -\mathbf{A}^T$$

Orthogonal Matrix. A square matrix A is called an orthogonal matrix if the product of the matrix A and the transpose matrix A' is an identity matrix *e.g.*,

$$A \cdot A' = I$$

if $|A| = 1$, matrix A is proper.

Conjugate of a Matrix

Let
$$A = \begin{bmatrix} 1+i & 2-3i & 4 \\ 7+2i & -i & 3-2i \end{bmatrix}$$

Conjugate of matrix A is $\bar{A}$

$$\bar{A} = \begin{bmatrix} 1-i & 2+3i & 4 \\ 7-2i & i & 3+2i \end{bmatrix}$$

Matrix A^θ . Transpose of the conjugate of a matrix A is denoted by A^θ .

Let
$$A = \begin{bmatrix} 1+i & 2-3i & 4 \\ 7+2i & -i & 3-2i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 1-i & 2+3i & 4 \\ 7-2i & +i & 3+2i \end{bmatrix}$$

$$(\bar{A})' = \begin{bmatrix} 1-i & 7-2i \\ 2+3i & i \\ 4 & 3+2i \end{bmatrix}$$

$$A^\theta = \begin{bmatrix} 1-i & 7-2i \\ 2+3i & i \\ 4 & 3+2i \end{bmatrix}$$

(o) **Unitary Matrix.** A square matrix A is said to be unitary if

$$A^\theta A = I$$

e.g. $A = \begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix}, \quad A^\theta = \begin{bmatrix} \frac{1-i}{2} & \frac{1-i}{2} \\ \frac{-1-i}{2} & \frac{1+i}{2} \end{bmatrix}, \quad A \cdot A^\theta = I$

(p) **Hermitian Matrix.** A square matrix $A = (a_{ij})$ is called Hermitian matrix, if every i - j th element of A is equal to conjugate complex j - i th element of A .

In other words, $a_{ij} = \bar{a}_{ji}$

e.g. $\begin{bmatrix} 1 & 2+3i & 3+i \\ 2-3i & 2 & 1-2i \\ 3-i & 1+2i & 5 \end{bmatrix}$

Necessary and sufficient condition for a matrix A to be Hermitian is that $A = A^\theta$ i.e. conjugate transpose of A

$$\Rightarrow A = (\bar{A})'.$$

(q) **Skew Hermitian Matrix.** A square matrix $A = (a_{ij})$ will be called a Skew Hermitian matrix if every i - j th element of A is equal to negative conjugate complex of j - i th element of A .

In other words,
$$a_{ij} = -\bar{a}_{ji}$$

All the elements in the principal diagonal will be of the form

$$a_{ii} = -\bar{a}_{ii} \quad \text{or} \quad a_{ii} + \bar{a}_{ii} = 0$$

$$\text{If } a_{ii} = a + ib \quad \text{then } \bar{a}_{ii} = a - ib$$

$$(a + ib) + (a - ib) = 0 \quad \Rightarrow \quad 2a = 0 \Rightarrow a = 0$$

So, a_{ii} is pure imaginary $\Rightarrow a_{ii} = 0$.

Hence, all the diagonal elements of a Skew Hermitian Matrix are either zeros or pure imaginary.

e.g.
$$\begin{bmatrix} i & 2-3i & 4+5i \\ -(2+3i) & 0 & 2i \\ -(4-5i) & 2i & -3i \end{bmatrix}$$

The necessary and sufficient condition for a matrix A to be Skew Hermitian is that

$$A^\theta = -A$$

$$(\bar{A})' = -A$$

(r) **Idempotent Matrix.** A matrix, such that $A^2 = A$ is called Idempotent Matrix.

$$\text{e.g. } A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}, A^2 = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = A$$

(s) **Periodic Matrix.** A matrix A will be called a Periodic Matrix, if

$$A^{k+1} = A$$

where k is a +ve integer. If k is the least +ve integer, for which $A^{k+1} = A$, then k is said to be the period of A . If we choose $k = 1$, we get $A^2 = A$ and we call it to be idempotent matrix.

(t) **Nilpotent Matrix.** A matrix will be called a Nilpotent matrix, if $A^k = 0$ (null matrix) where k is a +ve integer ; if however k is the least +ve integer for which $A^k = 0$, then k is the *index* of the nilpotent matrix.

$$\text{e.g., } A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}, A^2 = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix} \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

A is nilpotent matrix whose index is 2.

(u) **Involuntary Matrix.** A matrix A will be called an Involuntary matrix, if $A^2 = I$ (unit matrix). Since $I^2 = I$ always $\therefore$ Unit matrix is involuntary.

(v) **Equal Matrices.** Two matrices are said to be equal if

(i) They are of the same order.

(ii) The elements in the corresponding positions are equal.

Thus if

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

Here

$$A = B$$

(w) **Singular Matrix.** If the determinant of the matrix is zero, then the matrix is known as

singular matrix e.g. $A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ is singular matrix, because $|A| = 6 - 6 = 0$.

Example 1. Find the values of x , y , z and ' a ' which satisfy the matrix equation.

$$\begin{bmatrix} x+3 & 2y+x \\ z-1 & 4a-6 \end{bmatrix} = \begin{bmatrix} 0 & -7 \\ 3 & 2a \end{bmatrix}$$

Solution. As the given matrices are equal, so their corresponding elements are equal.

$$x+3=0 \quad \Rightarrow \quad x=-3 \quad \dots(1)$$

$$2y+x=-7 \quad \dots(2)$$

$$z-1=3 \quad \Rightarrow \quad z=4 \quad \dots(3)$$

$$4a-6=2a \quad \Rightarrow \quad a=3 \quad \dots(4)$$

Putting the value of $x=-3$ from (1) into (2), we have

$$2y-3=-7 \quad \Rightarrow \quad y=-2$$

Hence, $x=-3, y=-2, z=4, a=3$

Ans.

PROPERTIES OF MATRIX MULTIPLICATION

1. Multiplication of matrices is not commutative.

$$AB \neq BA$$

2. Matrix multiplication is associative, if conformability is assured.

$$A (BC) = (AB) C$$

3. Matrix multiplication is distributive with respect to addition.

$$A (B + C) = AB + AC$$

4. Multiplication of matrix A by unit matrix.

$$AI = IA = A$$

5. Multiplicative inverse of a matrix exists if $|A| \neq 0$.

$$A \cdot A^{-1} = A^{-1} \cdot A = I$$

6. If A is a square then $A \times A = A^2$, $A \times A \times A = A^3$.

7. $A^0 = I$

8. $I^n = I$, where n is positive integer.

Example 5. If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 0 \\ 2 & -1 \end{bmatrix}$

obtain the product AB and explain why BA is not defined.

Solution. The number of columns in A is 3 and the number of rows in B is also 3, therefore the product AB is defined.

* TO COMPUTE THE INVERSE OF A MATRIX FROM ELEMENTARY MATRICES (Gauss-jordan Method)

If A is reduced to I by elementary transformation then

$$PA = I \quad \text{where} \quad P = P_n P_{n-1} \cdots P_2 P_1$$

$$\therefore P = A^{-1} \quad = \text{Elementary matrix.}$$

Working rule. Write $A = IA$. Perform elementary row transformation on A of the left side and on I of the right hand side so that A is reduced to I and I of right hand side is reduced to P getting $I = PA$.

Then P is the inverse of A .

* THE INVERSE OF A SYMMETRIC MATRIX

The elementary transformations are to be transformed so that the property of being symmetric is preserved. This requires that the transformations occur in pairs, a row transformation must be followed immediately by the same column transformation.

Example 1. Find the inverse of the following matrix employing elementary transformations:

$$\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

(U.P., I Semester, Compartment 2002)

Solution. The given matrix is $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$

$$\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \Rightarrow \begin{bmatrix} 1 & -1 & \frac{4}{3} \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} R_1 \rightarrow \frac{R_1}{3} \\ A \end{matrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & \frac{4}{3} \\ 0 & -1 & \frac{4}{3} \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ -\frac{2}{3} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} A \\ R_2 \rightarrow R_2 - 2R_1 \end{matrix} \Rightarrow \begin{bmatrix} 1 & -1 & \frac{4}{3} \\ 0 & 1 & -\frac{4}{3} \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ \frac{2}{3} & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} A \\ R_2 \rightarrow -R_2 \end{matrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & \frac{4}{3} \\ 0 & 1 & -\frac{4}{3} \\ 0 & 0 & -\frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ \frac{2}{3} & -1 & 0 \\ \frac{2}{3} & -1 & 1 \end{bmatrix} A \quad R_3 \rightarrow R_3 + R_2 \quad \Rightarrow \begin{bmatrix} 1 & -1 & \frac{4}{3} \\ 0 & 1 & -\frac{4}{3} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ \frac{2}{3} & -1 & 0 \\ -2 & 3 & -3 \end{bmatrix} A \quad R_3 \rightarrow -3R_3$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -4 & 4 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix} A \quad \begin{array}{l} R_1 \rightarrow R_1 - \frac{4}{3}R_3 \\ R_2 \rightarrow R_2 + \frac{4}{3}R_3 \end{array}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix} A \quad R_1 \rightarrow R_1 + R_2$$

$$\text{Hence, } A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

Ans.

Example 2. Find by elementary row transformation the inverse of the matrix.

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

(U.P., I Semester, Winter 2003, 2000)

Solution. Let $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$

Elementary row transformation, which will reduce $A = IA$ to $I = PA$, then matrix P will be the inverse of matrix A .

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} R_1 \leftrightarrow R_2 \\ A \end{matrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -5 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & 1 \end{bmatrix} A \begin{matrix} R_3 \rightarrow R_3 - 3R_1 \\ \end{matrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 5 & -3 & 1 \end{bmatrix} A \begin{matrix} R_3 \rightarrow R_3 + 5R_2 \\ \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix} \begin{matrix} A \\ \\ R_3 \rightarrow \frac{1}{2}R_3 \end{matrix} \Rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{15}{2} & \frac{11}{2} & -\frac{3}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix} \begin{matrix} R_1 \rightarrow R_1 - 3R_3 \\ A \\ R_2 \rightarrow R_2 - 2R_3, \end{matrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix} \begin{matrix} R_1 \rightarrow R_1 - 2R_2 \\ A \\ \end{matrix}$$

$$I = PA \Rightarrow P = A^{-1}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

Ans.

Example 3. Find the inverse of the matrix M by applying elementary transformations

$$\begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix}.$$

[U.P.T.U.(C.O.) 2003]

Solution. Here, we have $A = \begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix}$

Let $\begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix} \xrightarrow{I_1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} A$

$$\begin{bmatrix} 1 & 1 & -1 & -2 \\ 0 & 2 & 1 & 3 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix} \xrightarrow{II} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} R_1 \leftrightarrow R_2 \\ A \end{matrix}$$

$$\begin{bmatrix} 1 & 1 & -1 & -2 \\ 0 & 2 & 1 & 3 \\ 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{array}{l} A \\ R_3 \rightarrow R_3 - R_1 \\ R_4 \rightarrow R_4 + R_1 \end{array}$$

$$\begin{bmatrix} 1 & 1 & -1 & -2 \\ 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 3 \\ 0 & 2 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{array}{l} A \\ R_3 \leftrightarrow R_2 \end{array}$$

$$\begin{bmatrix} 1 & 1 & -1 & -2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -1 & -3 \\ 0 & 0 & -1 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & 2 & -2 & 0 \\ 0 & 3 & -2 & 1 \end{bmatrix} \begin{array}{l} A \\ R_3 \rightarrow R_3 - 2R_2 \\ R_4 \rightarrow R_4 - 2R_2 \end{array}$$

$$\begin{bmatrix} 1 & 1 & -1 & -2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & 2 & -2 & 0 \\ -1 & 1 & 0 & 1 \end{bmatrix} \begin{array}{l} A \\ R_4 \rightarrow R_4 - R_3 \end{array}$$

$$\begin{bmatrix} 1 & 1 & -1 & -2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & -2 & 2 & 0 \\ -1 & 1 & 0 & 1 \end{bmatrix} \begin{array}{l} A \\ R_3 \rightarrow -R_3 \end{array}$$

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 3 & 0 & 2 \\ 3 & -4 & 1 & -3 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{bmatrix} \begin{array}{l} R_1 \rightarrow R_1 + 2R_4 \\ R_2 \rightarrow R_2 - 3R_4 \\ R_3 \rightarrow R_3 - 3R_4 \\ A \end{array}$$

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -2 & 2 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{bmatrix} \begin{array}{l} R_1 \rightarrow R_1 + R_3 \\ R_2 \rightarrow R_2 - R_3 \\ A \end{array}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{bmatrix} \begin{array}{l} R_1 \rightarrow R_1 - R_2 \\ A \end{array}$$

$$I = A^{-1}A$$

$$A^{-1} = \begin{bmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{bmatrix}$$

Hence,

Ans.

EXERCISE

Find the inverse of the following matrices:

1. $\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

Ans. $\begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

2. $\begin{bmatrix} 1 & -1 & 1 \\ 4 & 1 & 0 \\ 8 & 1 & 1 \end{bmatrix}$

Ans. $\begin{bmatrix} 1 & 2 & -1 \\ -4 & -7 & 4 \\ -4 & -9 & 5 \end{bmatrix}$

3. Use elementary row operations to find inverse of $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$

Ans. $\frac{1}{4} \begin{bmatrix} 12 & 4 & 6 \\ -5 & -1 & -3 \\ -1 & -1 & -1 \end{bmatrix}$

(AMETE, June 2010)

4. $\begin{bmatrix} 2 & 1 & -1 & 2 \\ 1 & 3 & 2 & -3 \\ -1 & 2 & 1 & -1 \\ 2 & -3 & -1 & 4 \end{bmatrix}$

Ans. $\frac{1}{18} \begin{bmatrix} 2 & 5 & -7 & 1 \\ 5 & -1 & 5 & -2 \\ -7 & 5 & 11 & 10 \\ 1 & -2 & 10 & 5 \end{bmatrix}$

5. $\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 3 & 2 \\ 2 & 4 & 3 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix}$ *(Q. Bank U.P. II Semester 2001)*

Ans. $\begin{bmatrix} 1 & -2 & 1 & 0 \\ 1 & -2 & 2 & -3 \\ 0 & 1 & -1 & 1 \\ -2 & 3 & -2 & 3 \end{bmatrix}$

RANK OF MATRIX

11.4 RANK OF A MATRIX

The rank of a matrix is said to be r if

- (a) It has at least one non-zero minor of order r .
- (b) Every minor of A of order higher than r is zero.

Note: (i) Non-zero row is that row in which all the elements are not zero.

(ii) The rank of the product matrix AB of two matrices A and B is less than the rank of either of the matrices A and B .

(iii) Corresponding to every matrix A of rank r , there exist non-singular matrices P and Q such

that
$$PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

11.5 NORMAL FORM (CANONICAL FORM)

By performing elementary transformation, any non-zero matrix A can be reduced to one of the following four forms, called the Normal form of A :

$$(i) I_r \qquad (ii) [I_r \ 0] \qquad (iii) \begin{bmatrix} I_r \\ 0 \end{bmatrix} \qquad (iv) \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

The number r so obtained is called the rank of A and we write $\rho(A) = r$. The form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ is called first canonical form of A . Since both row and column transformations may be used here, the element 1 of the first row obtained can be moved in the first column. Then both the first row and first column can be cleared of other non-zero elements. Similarly, the element 1 of the second row can be brought into the second column, and so on.

Example 1. Find the rank of the following matrix by reducing it to normal form –

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix} \quad (\text{U.P. I Sem., Com. 2002, Winter 2001})$$

Solution.

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 6 & -11 \\ 0 & -7 & 4 & -7 \\ 0 & 0 & 1 & -2 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 4R_1 \\ R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 6 & -11 \\ 0 & -7 & 4 & -7 \\ 0 & 0 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 6 & -11 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 1 & -2 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - R_2 \end{array}$$

$$C_2 \rightarrow C_2 - 2C_1, C_3 \rightarrow C_3 + C_1, C_4 \rightarrow C_4 - 3C_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 0 & 0 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 0 & 0 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} R_4 \rightarrow R_4 + \frac{1}{2}R_3 \end{array}$$

$$C_3 \rightarrow C_3 + \frac{6}{7}C_2, C_4 \rightarrow C_4 - \frac{11}{7}C_2,$$

$$C_4 \rightarrow C_4 + 2C_3 \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} R_2 \rightarrow -1/7 R_2 \\ R_3 \rightarrow -1/2 R_3 \end{array}$$

Rank of $A = 3$

Ans.

Example 2. For which value of 'b' the rank of the matrix

$$A = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ b & 13 & 10 \end{bmatrix} \text{ is 2,} \quad (\text{U.P., I Semester, 2008})$$

Solution. Here, we have

$$\begin{aligned} A &= \begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ b & 13 & 10 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ 0 & 13-5b & 10-4b \end{bmatrix} R_3 \rightarrow R_3 - bR_1 \\ &\sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 2 \\ 0 & 13-5b & 10-4b \end{bmatrix} \begin{array}{l} C_2 \rightarrow C_2 - 5C_1 \\ C_3 \rightarrow C_3 - 4C_1 \end{array} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 2 \\ 0 & 0 & \frac{2(2-b)}{3} \end{bmatrix} R_3 \rightarrow R_3 - \frac{13-5b}{3}R_2 \end{aligned}$$

If rank of A is 2, then $\frac{2(2-b)}{3}$ must be zero.

$$\text{i.e., } \frac{2(2-b)}{3} = 0 \quad \Rightarrow 2-b = 0 \quad \Rightarrow b = 2$$

Ans.

Example 3. Reduce the matrix to normal form and find its rank.

$$\begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 9 & 10 & 11 & 12 \end{bmatrix}$$

(R.G.P.V., Bhopal, I Sem. April 2009, 2003)

Solution.

$$\begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 9 & 10 & 11 & 12 \end{bmatrix} \sim \begin{bmatrix} 2 & 3 & 4 & 5 \\ 0 & -\frac{1}{2} & -1 & \frac{-3}{2} \\ 0 & -1 & -2 & -3 \\ 0 & \frac{-7}{2} & -7 & \frac{-21}{2} \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - \frac{3}{2}R_1 \\ R_3 \rightarrow R_3 - 2R_1 \\ R_4 \rightarrow R_4 - \frac{9}{2}R_1 \end{array}$$

$$\sim \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & -1 & \frac{-3}{2} \\ 0 & -1 & -2 & -3 \\ 0 & \frac{-7}{2} & -7 & \frac{-21}{2} \end{bmatrix} \begin{array}{l} C_2 \rightarrow C_2 - \frac{3}{2}C_1 \\ C_3 \rightarrow C_3 - 2C_1 \\ C_4 \rightarrow C_4 - \frac{5}{2}C_1 \end{array} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{bmatrix} \begin{array}{l} R_1 \rightarrow \frac{1}{2}R_1 \\ R_2 \rightarrow -2R_2 \\ R_3 \rightarrow -R_3 \\ R_4 \rightarrow \frac{-2}{7}R_4 \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2 \end{array} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} C_3 \rightarrow C_3 - 2C_2 \\ C_4 \rightarrow C_4 - 3C_2 \end{array} = \begin{bmatrix} I_2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence its rank = 2

Ans.

Example 4. Find the rank of the matrix.

$$A = \begin{bmatrix} 1 & 3 & 4 & 2 \\ 2 & -1 & 3 & 2 \\ 3 & -5 & 2 & 2 \\ 6 & -3 & 8 & 6 \end{bmatrix}, \text{ by reducing it to normal form.}$$

(Uttarakhand, I semester, Dec. 2006)

Solution. We have, $A = \begin{bmatrix} \textcircled{1} & 3 & 4 & 2 \\ 2 & -1 & 3 & 2 \\ 3 & -5 & 2 & 2 \\ 6 & -3 & 8 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 4 & 2 \\ 0 & -7 & -5 & -2 \\ 0 & -14 & -10 & -4 \\ 0 & -21 & -16 & -6 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 - 6R_1 \end{array}$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \textcircled{-7} & -5 & -2 \\ 0 & -14 & -10 & -4 \\ 0 & -21 & -16 & -6 \end{bmatrix} \begin{array}{l} C_2 \rightarrow C_2 - 3C_1 \\ C_3 \rightarrow C_3 - 4C_1 \\ C_4 \rightarrow C_4 - 2C_1 \end{array}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & -5 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - 2R_2 \\ R_4 \rightarrow R_4 - 3R_2 \end{array} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & -5 & -2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} R_3 \leftrightarrow R_4 \end{array}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} C_3 \rightarrow C_3 - \frac{5}{7}C_2 \\ C_4 \rightarrow C_4 - \frac{2}{7}C_2 \end{array} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} R_2 \rightarrow -\frac{1}{7}R_2 \\ R_3 \rightarrow -R_3 \end{array}$$

$$= \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix} \text{ which is normal form.}$$

Hence, Rank (A) = 3.

Ans.

Example 5. Reduce the matrix A to its normal form, when

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$$

Hence, find the rank of A .

(U.P., I Semester, Dec. 2004, Winter 2001)

RANK OF MATRIX BY TRIANGULAR FORM

Rank = Number of non-zero row in upper triangular matrix.

Note. Non-zero row is that row which does not contain all the elements as zero.

Example 8. Find the rank of the matrix

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix} \quad (U.P., I Semester, Winter 2003, 2000)$$

Solution.

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 1 & 1 & 3 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \\ \\ R_3 \rightarrow R_3 + R_2 \end{array}$$

Rank = Number of non zero rows = 2.

Ans.

Example 2. Find the rank of the matrix

$$\begin{bmatrix} -1 & 2 & 3 & -2 \\ 2 & -5 & 1 & 2 \\ 3 & -8 & 5 & 2 \\ 5 & -12 & -1 & 6 \end{bmatrix}$$

Solution.

$$\begin{bmatrix} -1 & 2 & 3 & -2 \\ 2 & -5 & 1 & 2 \\ 3 & -8 & 5 & 2 \\ 5 & -12 & -1 & 6 \end{bmatrix} \sim \begin{bmatrix} -1 & 2 & 3 & -2 \\ 0 & -1 & 7 & -2 \\ 0 & -2 & 14 & -4 \\ 0 & -2 & 14 & -4 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 + 2 R_1 \\ R_3 \rightarrow R_3 + 3 R_1 \\ R_4 \rightarrow R_4 + 5 R_1 \end{array}$$
$$\sim \begin{bmatrix} -1 & 2 & 3 & -2 \\ 0 & -1 & 7 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - 2 R_2 \\ R_4 \rightarrow R_4 - 2 R_2 \end{array}$$

Here the 4th order and 3rd order minors are zero. But a minor of second order

$$\begin{vmatrix} 3 & -2 \\ 7 & -2 \end{vmatrix} = -6 + 14 = 8 \neq 0$$

Rank = Number of non-zero rows = 2.

Ans.

Example. 3:- Use elementary transformation to reduce the following matrix A to triangular form and hence find the rank of A .

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

(R.G.P.V., Bhopal, June 2007, Winter 2003, U.P., I Semester, Dec. 2005)

Solution. We have,

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} \approx \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} R_1 \leftrightarrow R_2$$

$$\approx \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix} \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 - 6R_1 \end{matrix} \approx \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 0 & 33/5 & 22/5 \\ 0 & 0 & 33/5 & 22/5 \end{bmatrix} \begin{matrix} R_3 \rightarrow R_3 - 4/5 R_2 \\ R_4 \rightarrow R_4 - 9/5 R_2 \end{matrix}$$

$$\approx \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 0 & 33/5 & 22/5 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_4 \rightarrow R_4 - R_3$$

$R(A)$ = Number of non-zero rows.

$$\Rightarrow R(A) = 3$$

Ans.

EXERCISE 10.7

Find the rank of the following matrices:

1.
$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$$

Ans. 2

2.
$$\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & -3 \end{bmatrix}$$

Ans. 3

3.
$$\begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}$$

Ans. 2

4.
$$\begin{bmatrix} 2 & 4 & 3 & -2 \\ -3 & -2 & -1 & 4 \\ 6 & -1 & 7 & 2 \end{bmatrix}$$

Ans. 3

5.
$$\begin{bmatrix} 3 & 4 & 1 & 1 \\ 2 & 4 & 3 & 6 \\ -1 & -2 & 6 & 4 \\ 1 & -1 & 2 & -3 \end{bmatrix}$$

Ans. 4

6.
$$\begin{bmatrix} 1 & 4 & 3 & -2 & 1 \\ -2 & -3 & -1 & 4 & 3 \\ -1 & 6 & 7 & 2 & 9 \\ -3 & 3 & 6 & 6 & 12 \end{bmatrix}$$

Ans. 2

$$x - y + 2z = 3 \quad \dots(1)$$

$$3y + z = 2 \quad \dots(2)$$

$$\frac{-32}{3}z = \frac{-64}{3} \Rightarrow z = 2$$

Putting the value of z in (2), we get

$$3y + 2 = 2 \Rightarrow y = 0$$

Putting the value of y, z in (1), we get

$$x - 0 + 4 = 3 \Rightarrow x = -1$$

$$x = -1, y = 0, z = 2 \quad \text{Ans.}$$

Example 2. Find all the solutions of the system of equations

$$x_1 + 2x_2 - x_3 = 1, \quad 3x_1 - 2x_2 + 2x_3 = 2, \quad 7x_1 - 2x_2 + 3x_3 = 5$$

Solution.
$$\begin{bmatrix} 1 & 2 & -1 \\ 3 & -2 & 2 \\ 7 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1, R_3 \rightarrow R_3 - 7R_1$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -8 & 5 \\ 0 & -16 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 & 2 & -1 \\ 0 & -8 & 5 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} R_3 \rightarrow R_3 - 2R_2$$

$$x_1 + 2x_2 - x_3 = 1 \quad \dots(1)$$

$$-8x_2 + 5x_3 = -1 \quad \dots(2)$$

Let $x_3 = k$

Putting $x_3 = k$ in (2), we get

$$-8x_2 + 5k = -1 \Rightarrow x_2 = \frac{1}{8}(5k + 1)$$

Substituting the values of x_3, x_1 in (1), we get

$$x_1 + \frac{1}{4}(5k + 1) - k = 1$$

$$\therefore x_1 = 1 + k - \frac{5k}{4} - \frac{1}{4} = -\frac{k}{4} + \frac{3}{4}$$

$$\therefore x_1 = -\frac{k}{4} + \frac{3}{4}, x_2 = \frac{5k}{8} + \frac{1}{8}, x_3 = k$$

The equations have infinite solution.

Ans.

Example 3. Find the general solution of the system of equations:

$$\begin{aligned}3x_1 + 2x_3 + 2x_4 &= 0 \\ -x_1 + 7x_2 + 4x_3 + 9x_4 &= 0 \\ 7x_1 - 7x_2 - 5x_4 &= 0\end{aligned}$$

Solution. The system of equations in the matrix form is expressed as

$$\begin{bmatrix} 3 & 0 & 2 & 2 \\ -1 & 7 & 4 & 9 \\ 7 & -7 & 0 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 7 & 4 & 9 \\ 3 & 0 & 2 & 2 \\ 7 & -7 & 0 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} -1 & 7 & 4 & 9 \\ 0 & 21 & 14 & 29 \\ 0 & 42 & 28 & 58 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 + 3R_1 \\ R_3 \rightarrow R_3 + 7R_1 \end{array}$$

$$\begin{bmatrix} -1 & 7 & 4 & 9 \\ 0 & 21 & 14 & 29 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} R_3 \rightarrow R_3 - 2R_2$$

$$-x_1 + 7x_2 + 4x_3 + 9x_4 = 0 \quad \dots(1)$$

$$21x_2 + 14x_3 + 29x_4 = 0 \quad \dots(2)$$

Let $x_4 = a, x_3 = b$

From (2), $21x_2 + 14b + 29a = 0$ or $x_2 = -\frac{2b}{3} - \frac{29a}{21}$

From (1), $-x_1 + 7\left(-\frac{2b}{3} - \frac{29a}{21}\right) + 4b + 9a = 0$

$$x_1 = -\frac{2a}{3} - \frac{2b}{3}$$

$$x_1 = -\frac{2}{3}(a+b), x_2 = -\frac{1}{21}(29a+14b)$$

$$x_3 = b, x_4 = a$$

Ans.

LINEAR DEPENDENCE AND INDEPENDENCE OF VECTORS

Vectors (matrices) $X_1, X_2, \dots, X_n$ are said to be dependent if

(1) all the vectors (row or column matrices) are of the same order.

(2) n scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ (not all zero) exist such that

$$\lambda_1 X_1 + \lambda_2 X_2 + \lambda_3 X_3 + \dots + \lambda_n X_n = 0$$

Otherwise they are linearly independent.

Remember: If in a set of vectors, any vector of the set is the combination of the remaining vectors, then the vectors are called dependent vectors.

Example 1.1. *Examine the following vectors for linear dependence and find the relation if it exists.*

$$X_1 = (1, 2, 4), X_2 = (2, -1, 3), X_3 = (0, 1, 2), X_4 = (-3, 7, 2) \quad (U.P., I Sem. Winter 2002)$$

Solution. Consider the matrix equation

$$\lambda_1 X_1 + \lambda_2 X_2 + \lambda_3 X_3 + \lambda_4 X_4 = 0$$

$$\Rightarrow \lambda_1 (1, 2, 4) + \lambda_2 (2, -1, 3) + \lambda_3 (0, 1, 2) + \lambda_4 (-3, 7, 2) = 0$$

$$\lambda_1 + 2\lambda_2 + 0\lambda_3 - 3\lambda_4 = 0$$

$$2\lambda_1 - \lambda_2 + \lambda_3 + 7\lambda_4 = 0$$

$$4\lambda_1 + 3\lambda_2 + 2\lambda_3 + 2\lambda_4 = 0$$

This is the homogeneous system

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 2 & -1 & 1 & 7 \\ 4 & 3 & 2 & 2 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ or } A \lambda = 0$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & -5 & 2 & 14 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - 2 R_1 \\ R_3 \rightarrow R_3 - 4 R_1 \end{array}$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$$

$$\lambda_1 + 2 \lambda_2 - 3 \lambda_4 = 0$$

$$-5 \lambda_2 + \lambda_3 + 13 \lambda_4 = 0$$

$$\lambda_3 + \lambda_4 = 0$$

Let $\lambda_4 = t, \lambda_3 + t = 0, \lambda_3 = -t$

$$-5 \lambda_2 - t + 13 t = 0, \lambda_2 = \frac{12 t}{5}$$

$$\lambda_1 + \frac{24 t}{5} - 3 t = 0 \text{ or } \lambda_1 = \frac{-9 t}{5}$$

Hence, the given vectors are linearly dependent.

Substituting the values of λ in (1), we get

$$-\frac{9 t X_1}{5} + \frac{12 t}{5} X_2 - t X_3 + t X_4 = 0 \Rightarrow -\frac{9 X_1}{5} + \frac{12 X_2}{5} - X_3 + X_4 = 0$$

$$\Rightarrow 9 X_1 - 12 X_2 + 5 X_3 - 5 X_4 = 0$$

Ans.

Example 2. Define linear dependence and independence of vectors.

Examine for linear dependence $[1, 0, 2, 1]$, $[3, 1, 2, 1]$, $[4, 6, 2, -4]$, $[-6, 0, -3, -4]$ and find the relation between them, if possible.

Solution. Consider the matrix equation

$$\lambda_1 X_1 + \lambda_2 X_2 + \lambda_3 X_3 + \lambda_4 X_4 = 0 \quad \dots(1)$$

$$\lambda_1 (1, 0, 2, 1) + \lambda_2 (3, 1, 2, 1) + \lambda_3 (4, 6, 2, -4) + \lambda_4 (-6, 0, -3, -4) = 0$$

$$\lambda_1 + 3 \lambda_2 + 4 \lambda_3 - 6 \lambda_4 = 0$$

$$0 \lambda_1 + \lambda_2 + 6 \lambda_3 + 0 \lambda_4 = 0$$

$$2 \lambda_1 + 2 \lambda_2 + 2 \lambda_3 - 3 \lambda_4 = 0$$

$$\lambda_1 + \lambda_2 - 4 \lambda_3 - 4 \lambda_4 = 0$$

$$\begin{bmatrix} 1 & 3 & 4 & -6 \\ 0 & 1 & 6 & 0 \\ 2 & 2 & 2 & -3 \\ 1 & 1 & -4 & -4 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 4 & -6 \\ 0 & 1 & 6 & 0 \\ 0 & -4 & -6 & 9 \\ 0 & -2 & -8 & 2 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} R_3 \rightarrow R_3 - 2 R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array}$$

$$\begin{bmatrix} 1 & 3 & 4 & -6 \\ 0 & 1 & 6 & 0 \\ 0 & 0 & 18 & 9 \\ 0 & 0 & 4 & 2 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} R_3 \rightarrow R_3 + 4 R_2 \\ R_4 \rightarrow R_4 + 2 R_2 \end{array}$$

$$\begin{bmatrix} 1 & 3 & 4 & -6 \\ 0 & 1 & 6 & 0 \\ 0 & 0 & 18 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad R_4 \rightarrow R_4 - \frac{2}{9} R_3$$

$$\begin{aligned} \lambda_1 + 3 \lambda_2 + 4 \lambda_3 - 6 \lambda_4 &= 0 \\ \lambda_2 + 6 \lambda_3 &= 0 \\ 18 \lambda_3 + 9 \lambda_4 &= 0 \end{aligned}$$

$$\begin{aligned} \text{Let } \lambda_4 &= t, & 18 \lambda_3 + 9 t &= 0 \text{ or } \lambda_3 = \frac{-t}{2} \\ & \lambda_2 - 3 t &= 0 \text{ or } \lambda_2 = 3 t \\ & \lambda_1 + 9 t - 2t - 6t &= 0 \\ & \lambda_1 &= -t \end{aligned}$$

Substituting the values of λ_1 , λ_2 , λ_3 and λ_4 in (1), we get

$$-t X_1 + 3 t X_2 - \frac{t}{2} X_3 + t X_4 = 0 \text{ or } 2 X_1 - 6 X_2 + X_3 - 2 X_4 = 0$$

Ans.

Example 3. Show that row vectors of the matrix

$$\begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix} \text{ are linearly independent.}$$

(U.P., I Sem, Dec 2009)

Solution. Here, we have three vectors

$$X_1 = (1, 2, -2)'$$

$$X_2 = (-1, 3, 0)'$$

$$X_3 = (0, -2, 1)'$$

$$X_1 = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, X_2 = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}, X_3 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

Consider the equation

$$\lambda_1 X_1 + \lambda_2 X_2 + \lambda_3 X_3 = 0 \quad \dots(1)$$

$$\lambda_1 \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} + \lambda_2 \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix} + \lambda_3 \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda_1 - \lambda_2 + 0 \lambda_3 = 0$$

$$2 \lambda_1 + 3 \lambda_2 - 2 \lambda_3 = 0$$

$$-2 \lambda_1 + 0 \lambda_2 + \lambda_3 = 0$$

which is the system of homogeneous equations

$$\begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & -2 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 5 & -2 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 + 2R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 5 & -2 \\ 0 & 0 & \frac{1}{5} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \begin{matrix} \\ \\ R_3 \rightarrow R_3 + \frac{2}{5}R_2 \end{matrix}$$

$$\lambda_1 - \lambda_2 = 0 \quad \dots(2)$$

$$5\lambda_2 - 2\lambda_3 = 0 \quad \dots(3)$$

$$\frac{1}{5}\lambda_3 = 0 \Rightarrow \lambda_3 \quad \dots(4)$$

Putting the value of λ_3 in (3), we get

$$5\lambda_2 - 2(0) = 0 \Rightarrow \lambda_2 = 0$$

Putting the value of λ_2 in (2), we get

$$\lambda_1 - 0 = 0 \Rightarrow \lambda_1 = 0$$

Thus non zero values of $\lambda_1, \lambda_2, \lambda_3$ do not exist which can satisfy (1). Hence by definition the given system of vectors is linearly independent. **Proved.**

LINEARLY DEPENDENCE AND INDEPENDENCE OF VECTORS BY RANK METHOD

1. If the rank of the matrix of the given vectors is equal to number of vectors, then the vectors are linearly independent.
2. If the rank of the matrix of the given vectors is less than the number of vectors, then the vectors are linearly dependent.

Example | . Show using a matrix that the set of vectors

$X = [1, 2, -3, 4], Y = [3, -1, 2, 1], Z = [1, -5, 8, -7]$ is linearly dependent.

Solution. Here, we have

$$X = [1, 2, -3, 4], \quad Y = [3, -1, 2, 1], \quad Z = [1, -5, 8, -7]$$

Let us form a matrix of the above vectors

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 3 & -1 & 2 & 1 \\ 1 & -5 & 8 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 11 & -11 \\ 0 & -7 & 11 & -11 \end{bmatrix} \begin{matrix} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - R_1 \end{matrix} \sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 11 & -11 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ R_3 \rightarrow R_3 - R_2 \end{matrix}$$

Here the rank of the matrix = 2 < Number of vectors

Hence, vectors are linearly dependent.

Proved.

Example 2: Show using a matrix that the set of vectors : $[2, 5, 2, -3]$, $[3, 6, 5, 2]$, $[4, 5, 14, 14]$, $[5, 10, 8, 4]$ is linearly independent.

Solution. Here, the given vectors are

$$[2, 5, 2, -3], [3, 6, 5, 2], [4, 5, 14, 14], [5, 10, 8, 4]$$

Let us form a matrix of the above vectors :

$$\begin{aligned} & \begin{bmatrix} 2 & 5 & 2 & -3 \\ 3 & 6 & 5 & 2 \\ 4 & 5 & 14 & 14 \\ 5 & 10 & 8 & 4 \end{bmatrix} \sim \begin{bmatrix} 2 & 5 & 2 & -3 \\ 1 & 1 & 3 & 5 \\ 1 & -1 & 9 & 12 \\ 1 & 5 & -6 & -10 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_3 \end{array} \\ & \sim \begin{bmatrix} 1 & 1 & 3 & 5 \\ 2 & 5 & 2 & -3 \\ 1 & -1 & 9 & 12 \\ 1 & 5 & -6 & -10 \end{bmatrix} \begin{array}{l} R_1 \leftrightarrow R_2 \\ R_2 \leftrightarrow R_1 \end{array} \sim \begin{bmatrix} 1 & 1 & 3 & 5 \\ 0 & 3 & -4 & -13 \\ 0 & -2 & 6 & 7 \\ 0 & 4 & -9 & -15 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array} \\ & \sim \begin{bmatrix} 1 & 1 & 3 & 5 \\ 0 & 3 & -4 & -13 \\ 0 & 0 & \frac{10}{3} & \frac{-5}{3} \\ 0 & 0 & \frac{-11}{3} & \frac{7}{3} \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 + \frac{2}{3}R_2 \\ R_4 \rightarrow R_4 - \frac{4}{3}R_2 \end{array} \sim \begin{bmatrix} 1 & 1 & 3 & 5 \\ 0 & 3 & -4 & -13 \\ 0 & 0 & \frac{10}{3} & \frac{-5}{3} \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix} \begin{array}{l} R_4 \rightarrow R_4 + \frac{11}{10}R_3 \end{array} \end{aligned}$$

Here, the rank of the matrix = 4 = Number of vectors

Hence, the vectors are linearly independent.

Proved.

EXERCISE 10.

Examine the following system of vectors for linear dependence. If dependent, find the relation between them.

1. $X_1 = (1, -1, 1), X_2 = (2, 1, 1), X_3 = (3, 0, 2).$ **Ans.** Dependent, $X_1 + X_2 - X_3 = 0$
2. $X_1 = (1, 2, 3), X_2 = (2, -2, 6).$ **Ans.** Independent
3. $X_1 = (3, 1, -4), X_2 = (2, 2, -3), X_3 = (0, -4, 1).$ **Ans.** Dependent, $2 X_1 - 3 X_2 - X_3 = 0$
4. $X_1 = (1, 1, 1, 3), X_2 = (1, 2, 3, 4), X_3 = (2, 3, 4, 7).$ **Ans.** Dependent, $X_1 + X_2 - X_3 = 0$
5. $X_1 = (1, 1, -1, 1), X_2 = (1, -1, 2, -1), X_3 = (3, 1, 0, 1).$ **Ans.** Dependent, $2 X_1 + X_2 - X_3 = 0$
6. $X_1 = (1, -1, 2, 0), X_2 = (2, 1, 1, 1), X_3 = (3, -1, 2, -1), X_4 = (3, 0, 3, 1).$ **Ans.** Dependent, $X_1 + X_2 - X_4 = 0$

EIGEN VALUES, EIGEN VECTOR, CAYLEY HAMILTON THEOREM

INTRODUCTION

Eigen values and eigen vectors are used in the study of ordinary differential equations, analysing population growth and finding powers of matrices.

2.1.1 EIGEN VALUES

$$\text{Let } \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix}$$
$$AX = Y \quad \dots(1)$$

Where A is the matrix, X is the column vector and Y is also column vector.

Here column vector X is transformed into the column vector Y by means of the square matrix A .

Let X be a such vector which transforms into λX by means of the transformation (1). Suppose the linear transformation $Y = AX$ transforms X into a scalar multiple of itself i.e. λX .

$$\begin{aligned} AX &= Y = \lambda X \\ AX - \lambda IX &= 0 \\ (A - \lambda I) X &= 0 \end{aligned} \quad \dots(2)$$

Thus the unknown scalar λ is known as an eigen value of the matrix A and the corresponding non zero vector X as **eigen vector**.

The eigen values are also called characteristic values or proper values or latent values.

Let $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{bmatrix} \quad \text{characteristic matrix}$$

(b) **Characteristic Polynomial:** The determinant $|A - \lambda I|$ when expanded will give a polynomial, which we call as characteristic polynomial of matrix A .

For example;
$$\begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix}$$

$$= (2-\lambda)(6-5\lambda+\lambda^2-2) - 2(2-\lambda-1) + 1(2-3+\lambda)$$

$$= -\lambda^3 + 7\lambda^2 - 11\lambda + 5$$

- (c) **Characteristic Equation:** The equation $|A - \lambda I| = 0$ is called the characteristic equation of the matrix A e.g.

$$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$$

- (d) **Characteristic Roots or Eigen Values:** The roots of characteristic equation $|A - \lambda I| = 0$ are called characteristic roots of matrix A . e.g.

$$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 1)(\lambda - 5) = 0 \qquad \therefore \lambda = 1, 1, 5$$

Characteristic roots are 1, 1, 5.

Some Important Properties of Eigen Values

(AMIETE, Dec. 2009)

(1) Any square matrix A and its transpose A' have the same eigen values.

Note. The sum of the elements on the principal diagonal of a matrix is called the **trace** of the matrix.

(2) The sum of the eigen values of a matrix is equal to the **trace** of the matrix.

(3) The product of the eigen values of a matrix A is equal to the **determinant** of A .

(4) If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigen values of A , then the eigen values of

(i) kA are $k\lambda_1, k\lambda_2, \dots, k\lambda_n$ (ii) A^m are $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$

(iii) A^{-1} are $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}$.

(5) If λ be an eigen value of a non-singular matrix A then $\frac{|A|}{\lambda}$ is an eigen value of $\text{adj } A$.

(6) If A is singular then atleast one of its eigen value is zero.

Example 1. Find the characteristic roots of the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

Solution. The characteristic equation of the given matrix is

$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (6 - \lambda) (9 - 6\lambda + \lambda^2 - 1) + 2 (-6 + 2\lambda + 2) + 2(2 - 6 + 2\lambda) = 0$$

$$\Rightarrow -\lambda^3 + 12\lambda^2 - 36\lambda + 32 = 0$$

By trial, $\lambda = 2$ is a root of this equation.

$$\Rightarrow (\lambda - 2) (\lambda^2 - 10\lambda + 16) = 0 \Rightarrow (\lambda - 2) (\lambda - 2) (\lambda - 8) = 0$$

$$\Rightarrow \lambda = 2, 2, 8 \text{ are the characteristic roots or Eigen values.}$$

Ans.

Example 2. Find the eigen values of the matrix :

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

(R.G.P.V. Bhopal, I Semester, June 2007)

Solution. Let $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

The characteristic equation of A is

$$\begin{aligned} |A - \lambda I| &= 0 \\ \begin{vmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{vmatrix} &= 0 \end{aligned}$$

Expanding the determinant with the help of third row, we have

$$\Rightarrow (1-\lambda) \left[(2-\lambda)^2 - 1 \right] = 0 \quad \Rightarrow (1-\lambda)(\lambda^2 - 4\lambda + 4 - 1) = 0$$

$$\Rightarrow (1-\lambda)(\lambda^2 - 4\lambda + 3) = 0 \quad \Rightarrow (1-\lambda)(\lambda - 3)(\lambda - 1) = 0$$

$$\Rightarrow \lambda = 1, 1, 3$$

The eigen values of the given matrix are 1, 1 and 3.

Ans.

Example 3. The matrix A is defined as $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$

Find the eigen values of $3A^3 + 5A^2 - 6A + 2I$.

Solution. $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 & -3 \\ 0 & 3-\lambda & 2 \\ 0 & 0 & -2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1 - \lambda)(3 - \lambda)(-2 - \lambda) = 0 \text{ or } \lambda = 1, 3, -2$$

Eigen values of $A^3 = 1, 27, -8$; Eigen values of $A^2 = 1, 9, 4$

Eigen values of $A = 1, 3, -2$; Eigen values of $I = 1, 1, 1$

$\therefore$ Eigen values of $3A^3 + 5A^2 - 6A + 2I$

$$\text{First eigen value} = 3(1)^3 + 5(1)^2 - 6(1) + 2(1) = 4$$

$$\text{Second eigen value} = 3(27) + 5(9) - 6(3) + 2(1) = 110$$

$$\text{Third eigen value} = 3(-8) + 5(4) - 6(-2) + 2(1) = 10$$

Required eigen values are 4, 110, 10

Ans.

Example 4 : Find the eigen values and the corresponding eigen vectors of the matrix. [RTU, 2012]

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Solution Given matrix is

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Let λ be the eigen value of A.

The characteristic equation of A is given by $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} = 0$$

On expansion it gives

$$-\lambda^3 + 18\lambda^2 - 45\lambda = 0$$

$$\text{or } \lambda(\lambda - 3)(\lambda - 15) = 0$$

$$\text{or } \lambda = 0, 3, 15$$

$\therefore$ eigen values of A are 0, 3, 15

To find Eigen Vectors :

Let $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be the eigen vector correspond to λ .

$$\begin{aligned} \text{Then we have } & AX = \lambda X \\ \Rightarrow & (A - \lambda I) X = 0 \end{aligned}$$

$$\text{or } \begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad \dots(1)$$

Case (1) Put $\lambda = 0$ in equation (1) we get

$$\begin{aligned} 8x_1 - 6x_2 + 2x_3 &= 0 \\ -6x_1 + 7x_2 - 4x_3 &= 0 \\ 2x_1 - 4x_2 + 3x_3 &= 0 \end{aligned}$$

solving any two of the above equations by rule of cross multiplication, we have.

$$\frac{x_1}{24-14} = \frac{x_2}{-12+32} = \frac{x_3}{56-36}$$

$$\text{or } \frac{x_1}{10} = \frac{x_2}{20} = \frac{x_3}{20}$$

$$\text{or } \frac{x_1}{1} = \frac{x_2}{2} = \frac{x_3}{2} = k_1 \quad (k_1 \neq 0)$$

$$\Rightarrow X_1 = \begin{bmatrix} k_1 \\ 2k_1 \\ 2k_1 \end{bmatrix}$$

$\therefore$ Eigen vector corresponds to $\lambda = 0$ is $X_1 = \begin{bmatrix} k_1 \\ 2k_1 \\ 2k_1 \end{bmatrix}$ when $k_1 = 1$,

$$\text{we have } X_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Case (2) Put $\lambda = 3$ in equation (1) we get

$$\begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives,

$$\begin{aligned} 5x_1 - 6x_2 + 2x_3 &= 0 \\ -6x_1 + 4x_2 - 4x_3 &= 0 \\ 2x_1 - 4x_2 - 0x_3 &= 0. \end{aligned}$$

solving any two of the above equations, we get

$$\frac{x_1}{24-8} = \frac{x_2}{-12+20} = \frac{x_3}{20-36}$$

$$\text{or } \frac{x_1}{16} = \frac{x_2}{8} = \frac{x_3}{-16}$$

$$\text{or } \frac{x_1}{2} = \frac{x_2}{1} = \frac{x_3}{-2} = k_2 \quad (k_2 \neq 0)$$

$$\Rightarrow X_2 = \begin{bmatrix} 2k_2 \\ k_2 \\ -2k_2 \end{bmatrix}$$

$\therefore$ Eigen vector corresponds to $\lambda = 3$ is $X_2 = \begin{bmatrix} 2k_2 \\ k_2 \\ -2k_2 \end{bmatrix}$

when $k_2 = 1$, we have $X_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$

Case (3) Put $\lambda = 15$ in equation (1), we get

$$\begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives,

$$-7x_1 - 6x_2 + 2x_3 = 0$$

$$-6x_1 - 8x_2 - 4x_3 = 0$$

$$2x_1 - 4x_2 - 12x_3 = 0$$

solving any two of the above equations, we have

$$\frac{x_1}{24+16} = \frac{x_2}{-12-28} = \frac{x_3}{56-36}$$

$$\text{or } \frac{x_1}{40} = \frac{x_2}{-40} = \frac{x_3}{20}$$

$$\text{or } \frac{x_1}{2} = \frac{x_2}{-2} = \frac{x_3}{1} = k_3 \quad (k_3 \neq 0)$$

$$\Rightarrow X_3 = \begin{bmatrix} 2k_3 \\ -2k_3 \\ k_3 \end{bmatrix}$$

$\therefore$ Eigen vectors corresponding to $\lambda = 15$ is $X_3 = \begin{bmatrix} 2k_3 \\ -2k_3 \\ k_3 \end{bmatrix}$

Example 5 : Find the eigen values and eigen vectors of the matrix.

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

Solution Given matrix is

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

Let λ be the eigen value of A.

The characteristic equation of A is

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{vmatrix} = 0$$

On expanding along first row,

$$(-2-\lambda)[(1-\lambda)(-\lambda)-12] - 2[2(-\lambda)-6] - 3[-4+(1-\lambda)] = 0$$

$$\text{or } \lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

$$\text{or } (\lambda+3)(\lambda^2 - 2\lambda - 15) = 0$$

$$\text{or } (\lambda+3)(\lambda+3)(\lambda-5) = 0$$

$$\text{or } \lambda = -3, -3, 5$$

$\therefore$ Eigen values of A are $-3, -3, 5$.

To find Eigen Vectors :

Let $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be the eigen vector corresponding to λ . Then we have,

$$AX = \lambda X$$

$$\Rightarrow (A - \lambda I) X = 0$$

$$\text{or } \begin{bmatrix} -2 - \lambda & 2 & -3 \\ 2 & 1 - \lambda & -6 \\ -1 & -2 & 0 - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Case (1) Put $\lambda = -3$ in equation (1), we get

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives,

$$\begin{aligned} x_1 + 2x_2 - 3x_3 &= 0 \\ 2x_1 + 4x_2 - 6x_3 &= 0 \\ -x_1 - 2x_2 + 3x_3 &= 0 \end{aligned}$$

This is a set of one equation with three unknowns. Hence it has an infinite number of solutions. The above equation can be written as

$$x_1 + 2x_2 - 3x_3 = 0$$

By choosing,

$$x_2 = 0, x_1 = 3, x_3 = 1$$

and $x_3 = 0, x_1 = -2, x_2 = 1$ satisfy the above equation.

$\therefore$ One vector corresponding to $\lambda = -3$ is $\begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$ and the other vector corresponding to $\lambda = -3$ is $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$.

In this case, eigen vector corresponding to the root $\lambda = -3$ is the linear combination of the vectors $\begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$. Thus the eigen vector corresponding to $\lambda = -3$ is

$$k_1 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3k_1 - 2k_2 \\ k_2 \\ k_1 \end{bmatrix}$$

where k_1 and k_2 are constants.

Case (2) Put $\lambda = 5$ in equation (1), we have

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives,

$$-7x_1 + 2x_2 + 3x_3 = 0$$

$$2x_1 - 4x_2 - 6x_3 = 0$$

$$-x_1 - 2x_2 - 5x_3 = 0$$

solving any two of the above equations,

$$\frac{x_1}{-12-12} = \frac{x_2}{-6-42} = \frac{x_3}{28-4}$$

$$\text{or } \frac{x_1}{-24} = \frac{x_2}{-48} = \frac{x_3}{24}$$

$$\text{or } \frac{x_1}{-1} = \frac{x_2}{-2} = \frac{x_3}{1} = k_3$$

$$\therefore X_2 = \begin{bmatrix} -k_3 \\ -2k_3 \\ k_3 \end{bmatrix}$$

$\therefore$ Eigen vector corresponding to $\lambda = 5$ is $X_2 = \begin{bmatrix} -k_3 \\ -2k_3 \\ k_3 \end{bmatrix}$ put $k_3 = 1$

$\therefore$ Eigen vector is $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$.

Note : If X_1 and X_2 are two eigen vectors of a matrix A corresponding to same eigen value λ of A , then any linear combination of the form $k_1X_1 + k_2X_2$ is also gives eigen vector of A corresponding to the same eigen value λ .

Example $\hookrightarrow$: Find the eigen values and corresponding eigen vectors for the matrix

$$A = \begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

[RTU, 2009]

Solution Let λ be the eigen value of A . The characteristic equation of A is
 $(A - \lambda I) = 0$

$$\Rightarrow \begin{vmatrix} -1-\lambda & 2 & -2 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & 0-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (-1-\lambda) \begin{vmatrix} 2-\lambda & 1 \\ -1 & -\lambda \end{vmatrix} - 2 \begin{vmatrix} 1 & 1 \\ -1 & -\lambda \end{vmatrix} - 2 \begin{vmatrix} 1 & 2-\lambda \\ -1 & -1 \end{vmatrix} = 0$$

$$\Rightarrow -(1+\lambda)[-2\lambda + \lambda^2 + 1] - 2[-\lambda + 1] - 2[-1 + 2 - \lambda] = 0$$

$$\Rightarrow -\lambda^3 + \lambda^2 + 5\lambda - 5 = 0$$

$$\Rightarrow \lambda^3 - \lambda^2 - 5\lambda + 5 = 0$$

$$\Rightarrow \lambda = 1, \pm \sqrt{5}$$

$\therefore$ Eigen values of A are $1, \pm \sqrt{5}$

To find Eigen Vectors :

Let $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be the eigen vector corresponding to λ .

then, we have

$$(A - \lambda I) x = 0$$

$$\text{or } \begin{bmatrix} -1-\lambda & 2 & -2 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & 0-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

case-I : Put $\lambda = 1$

$$\begin{bmatrix} -2 & 2 & -2 \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} -x_1 + x_2 - x_3 = 0 \\ x_1 + x_2 + x_3 = 0 \end{cases} \Rightarrow x_1 = k_1, x_2 = 0, x_3 = -k_1$$

$$\therefore x_1 = \begin{bmatrix} k_1 \\ 0 \\ -k_1 \end{bmatrix}; k = 0, 1, 2, \dots$$

Case-II : Putting $\lambda = \pm\sqrt{5}$

$$\begin{bmatrix} -1 \mp \sqrt{5} & 2 & -2 \\ 1 & 2 \mp \sqrt{5} & 1 \\ -1 & -1 & \mp \sqrt{5} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow (-1 \mp \sqrt{5})x_1 + 2x_2 - 2x_3 = 0$$

$$x_1 + (2 \mp \sqrt{5})x_2 + x_3 = 0$$

$$-x_1 - x_2 \mp \sqrt{5}x_3 = 0$$

solving, we get eigen vector x_2 & x_3

Example 7 : Find the eigen values and eigen vectors of

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Solution Let λ be the eigen value of A.
The characteristic equation of A is,

$$|A - \lambda I| = 0$$

$$\text{or } \begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

On expanding along first row,

$$(6-\lambda)[(3-\lambda)(3-\lambda)-1] + 2[(-2)(3-\lambda)+2] + 2[2-2(3-\lambda)] = 0$$

$$\text{or } \lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

$$\text{or } (\lambda-2)(\lambda^2 - 10\lambda + 6) = 0$$

$$\text{or } (\lambda-2)(\lambda-2)(\lambda-8) = 0$$

$$\text{or } \lambda = 2, 2, 8$$

$\therefore$ Eigen values of A are 2, 2, 8.

To find Eigen Vectors :

Let $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be the eigen vector corresponding to λ . Then we have,

$$\Rightarrow \begin{aligned} AX &= \lambda X \\ (A - \lambda I) X &= 0 \end{aligned}$$

$$\text{or } \begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \dots(1)$$

Case (1) Put $\lambda = 2$ in equation (1), we have

$$\begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives,

$$\begin{aligned} 4x_1 - 2x_2 + 2x_3 &= 0 \\ -2x_1 + x_2 - x_3 &= 0 \\ 2x_1 - x_2 + x_3 &= 0 \end{aligned}$$

This is a set of one equation with three unknowns. Hence it has an infinite number of solutions. The above set can be written as

$$2x_1 - x_2 + x_3 = 0$$

Choosing $x_2 = 0, x_1 = 1, x_3 = -2$ and
 $x_3 = 0, x_1 = 1, x_2 = 2$ satisfies the equations.

$\therefore$ One eigen vector corresponding to $\lambda = 2$ is $\begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$ and the other vector for $\lambda = 2$ is $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$.

In this case, eigen vector corresponding to the root $\lambda = 2$ is the linear combination of the

vectors $\begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$. Thus, the eigen vector corresponding to $\lambda = 2$ is

$$k_1 \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} + k_2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} k_1 + k_2 \\ 2k_2 \\ -2k_1 \end{bmatrix} \text{ where } k_1 \text{ and } k_2 \text{ are constants.}$$

Case (2) Put $\lambda = 8$ in (1), we get

$$\begin{bmatrix} -2 & 2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives,

$$-2x_1 + 2x_2 + 2x_3 = 0$$

$$-2x_1 - 5x_2 - x_3 = 0$$

$$2x_1 - x_2 - 5x_3 = 0$$

Solving any two of the above equations, we get

$$\frac{x_1}{2+10} = \frac{x_2}{-4-2} = \frac{x_3}{10-4}$$

$$\alpha \frac{x_1}{12} = \frac{x_2}{-6} = \frac{x_3}{6}$$

$$\alpha \frac{x_1}{2} = \frac{x_2}{-1} = \frac{x_3}{1} = k_3 \text{ where } (k_3 \neq 0)$$

$$\therefore X_2 = \begin{bmatrix} 2k_3 \\ -k_3 \\ k_3 \end{bmatrix}$$

$\therefore$ Eigen vector corresponding to $\lambda = 8$ is $\begin{bmatrix} 2k_3 \\ -k_3 \\ k_3 \end{bmatrix}$. Put $k_3 = 1$ then eigen vector for $\lambda = 8$

$$\text{is } \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

Example 8 : Find the eigen values and eigen vector of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -4 & 7 \\ 0 & 0 & 7 \end{bmatrix}.$$

Solution Let λ be the eigen value of A.

The characteristic equation of A is,

$$|A - \lambda I| = 0$$

$$\text{or } \begin{vmatrix} 1-\lambda & 2 & 3 \\ 0 & -4-\lambda & 7 \\ 0 & 0 & 7-\lambda \end{vmatrix} = 0$$

On expanding along first row, we get

$$(1-\lambda)(-4-\lambda)(7-\lambda) = 0$$

$$\text{or } \lambda = 1, -4, 7$$

$\therefore$ Eigen values of A are 1, -4, 7.

To find Eigen Vectors :

Let $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be the eigen vector corresponding to λ . Then we have,

$$AX = \lambda X$$

$$\text{or } (A - \lambda I) X = 0$$

$$\text{or } \begin{bmatrix} 1-\lambda & 2 & 3 \\ 0 & -4-\lambda & 7 \\ 0 & 0 & 7-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Case (1) Put $\lambda = 1$ in equation (1), we get

$$\begin{bmatrix} 0 & 2 & 3 \\ 0 & -5 & 7 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives,

$$\begin{aligned} 2x_2 + 3x_3 &= 0 \\ -5x_2 + 7x_3 &= 0 \\ 6x_3 &= 0 \end{aligned}$$

$$\therefore x_3 = 0 \text{ and } x_2 = 0$$

since the above equation are not having x_1 , therefore we can't find x_1 hence x_1 is arbitrary

$$\therefore x_2 = x_3 = 0 \text{ and } x_1 = k_1, \text{ an arbitrary constant.}$$

$$\therefore \text{Eigen vector corresponding to } \lambda = 1 \text{ is } \begin{bmatrix} k_1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{put } k_1 = 1, \text{ we get } \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Case (2) Put $\lambda = -4$ in equation (1), we get

$$\begin{bmatrix} 5 & 2 & 3 \\ 0 & 0 & 7 \\ 0 & 0 & 11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives,

$$5x_1 + 2x_2 + 3x_3 = 0$$

$$7x_3 = 0$$

$$11x_3 = 0$$

Solving $x_3 = 0$ and $5x_1 + 2x_2 + 3x_3 = 0$ by cross multiplication, we have

$$\frac{x_1}{-2} = \frac{x_2}{5} = k_2 \quad (k_2 \neq 0)$$

$$\therefore X_2 = \begin{bmatrix} -2k_2 \\ 5k_2 \\ 0 \end{bmatrix}$$

$\therefore$ Eigen vector corresponding to $\lambda = -4$ is $\begin{bmatrix} -2k_2 \\ 5k_2 \\ 0 \end{bmatrix}$ put $k_2 = 1$ then eigen vector for $\lambda = -$

$$4 \text{ is } \begin{bmatrix} -2 \\ 5 \\ 0 \end{bmatrix}.$$

Case (3) Put $\lambda = 7$ in equation (1), we get

$$\begin{bmatrix} -6 & 2 & 3 \\ 0 & -11 & 7 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives,

$$\begin{aligned} -6x_1 + 2x_2 + 3x_3 &= 0 \\ -11x_2 + 7x_3 &= 0 \end{aligned}$$

On cross - multiplication, we get

$$\frac{x_1}{14+33} = \frac{x_2}{0+42} = \frac{x_3}{66-0}$$

or $\frac{x_1}{47} = \frac{x_2}{42} = \frac{x_3}{66} = k_3 \quad (k_3 \neq 0)$

$\therefore$ Eigen vector corresponding to $\lambda = 7$ is $\begin{bmatrix} 47k_3 \\ 42k_3 \\ 66k_3 \end{bmatrix}$. Put $k_3 = 1$ we get

$$X_3 = \begin{bmatrix} 47 \\ 42 \\ 66 \end{bmatrix}$$

Exercise

Q.1 Find the eigen values and eigen vectors of $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

Ans. $[2, 3, 6, \begin{bmatrix} k_1 \\ 0 \\ -k_1 \end{bmatrix}, \begin{bmatrix} k_2 \\ k_2 \\ k_2 \end{bmatrix}, \begin{bmatrix} k_3 \\ -2k_3 \\ k_3 \end{bmatrix}]$

Q.2 Find the eigen values and eigen vectors of matrix $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & 1 \\ 1 & -1 & 3 \end{bmatrix}$ [RTU, 2010]

Ans. $[1, 1, 4, \begin{bmatrix} k_1 \\ k_2 \\ 0 \end{bmatrix}, \begin{bmatrix} k_1 \\ 0 \\ k_2 \end{bmatrix}, \begin{bmatrix} k_1 \\ -k_2 \\ k_3 \end{bmatrix}]$

Q.3 Find the eigen values and eigen vectors of $A = \begin{bmatrix} -2 & 1 & 1 \\ -11 & 4 & 5 \\ -1 & 1 & 0 \end{bmatrix}$ [B.E., RTU, 2007]

Ans. $[1, 2, -1, \begin{bmatrix} k_1 \\ 2k_1 \\ k_1 \end{bmatrix}, \begin{bmatrix} k_2 \\ 3k_2 \\ k_2 \end{bmatrix}, \begin{bmatrix} 0 \\ -k_3 \\ k_3 \end{bmatrix}]$

Q.4 Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 9 & -1 & 9 \\ 3 & -1 & 3 \\ -7 & 1 & -7 \end{bmatrix}$

Ans. $[-1, 0, 2, \begin{bmatrix} k_1 \\ 0 \\ -k_1 \end{bmatrix}, \begin{bmatrix} k_2 \\ k_2 \\ -k_2 \end{bmatrix}, \begin{bmatrix} 4k_3 \\ k_3 \\ -3k_3 \end{bmatrix}]$

Vector space :- Motivation of the topic
come from (i) Vectors (ii) Matrices.

Let two vectors

$$\alpha = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}, \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}.$$

$$\text{Then } \alpha + \beta = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} + \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} \alpha_1 + \beta_1 \\ \alpha_2 + \beta_2 \end{bmatrix}$$

Also, let 'a' be a scalar. Then

$$a \cdot \alpha = a \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} a\alpha_1 \\ a\alpha_2 \end{bmatrix}$$

↓
Vector
Addition

↓
Scalar multiplication

Definition :- Let V be set of vectors.
and F be field (i.e: $\mathbb{R}$ = Set of real numbers).
 $\mathbb{C}$ = Set of complex numbers.

Two operation: vector addition & scalar multiplication.

i) Closure properties: $\forall \alpha, \beta \in V, \alpha + \beta \in V$
and $\forall q \in \mathbb{R}, \alpha \in V, q\alpha \in V$

The set V is closed under vector addition & scalar multiplication.

Vector addition satisfies the following axioms:

(a) Associative Property:- $\forall \alpha, \beta, \gamma \in V,$

$$\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma,$$

(b) Identity Element:- $\exists$ an unique 0 in V
such that $\alpha + 0 = 0 + \alpha = \alpha, \forall \alpha \in V.$

(c) Inverse Element:- $\exists \beta \in V$ such that
 $\alpha + \beta = \beta + \alpha = 0, \forall \alpha \in V$

(d) Commutative Law:- $\forall \alpha, \beta \in V, \alpha + \beta = \beta + \alpha$

ii) $\forall \alpha \in V, 1 \cdot \alpha = \alpha$

iii) $\forall a, b \in \mathbb{R}, \alpha \in V, (ab)\alpha = a(b\alpha)$

iv) $\forall a, b \in \mathbb{R}, \alpha \in V, (a+b)\alpha = a\alpha + b\alpha$

v) $\forall q \in \mathbb{R}, \alpha, \beta \in V, q(\alpha + \beta) = q\alpha + q\beta$

Ex. 1. $\mathbb{R}^2 = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mid x_1, x_2 \in \mathbb{R} \right\}$ and vector addition: component wise and Scalar multiplication: component wise.

$$x, y \in \mathbb{R}^2, \quad x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad x+y = \begin{pmatrix} x_1+y_1 \\ x_2+y_2 \end{pmatrix}$$

$$\text{Also, } ax = a \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} ax_1 \\ ax_2 \end{pmatrix}.$$

Closure property satisfies.

Because of commutativity & associativity of addition in $\mathbb{R}$, vector addition is also commutative & associative (a), (d).

Existence of zero vector $0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ ^{Scalars} from $\mathbb{R}$

Existence of inverse

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$\text{then } x+y = 0 = y+x$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1+y_1 \\ x_2+y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow x_1+y_1=0 \quad \& \quad x_2+y_2=0$$

$$\Rightarrow y_1 = -x_1 \quad \& \quad y_2 = -x_2$$

$$\Rightarrow y = \begin{pmatrix} -x_1 \\ -x_2 \end{pmatrix}$$

Also, V satisfies the all other properties. Hence $\mathbb{R}^2(\mathbb{R})$ is a vector space.

Singular Value Decomposition of a Matrix

Mathematically, the singular value decomposition of a matrix can be explained as follows:

Consider a matrix A of order $m \times n$.

This can be uniquely decomposed as:

$$A = UDV^T$$

U is $m \times n$ and column orthogonal (that means its columns are eigenvectors of AA^T)

$$(AA^T = UDV^T VDU^T = UD^2U^T)$$

V is $n \times n$ and orthogonal (that means its columns are eigenvectors of $A^T A$)

$$(A^T A = VDU^T UDV^T = VD^2V^T)$$

D is $n \times n$ diagonal, where non-negative real values are called singular values.

Learn how to find [eigenvalues](#) and [eigenvectors](#) of a matrix here.

Let $D = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_n)$ ordered such that $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$.

If σ is a singular value of A , its square is an eigenvalue of $A^T A$.

Also, let $U = (u_1 \ u_2 \ \dots \ u_n)$ and $V = (v_1 \ v_2 \ \dots \ v_n)$.

Therefore,

$$A = \sum_{i=1}^n \sigma_i u_i v_i^T$$

Here, the sum can be given from 1 to r so that r is the rank of matrix A .

Go through the example given below to understand the process of singular value decomposition of a matrix in a better way.

Question:

Find the singular value decomposition of a matrix

$$A = \begin{bmatrix} -4 & -7 \\ 1 & 4 \end{bmatrix}$$

.

Solution:

Given,

$$A = \begin{bmatrix} -4 & -7 \\ 1 & 4 \end{bmatrix}$$

So,

$$A^T = \begin{bmatrix} -4 & 1 \\ -7 & 4 \end{bmatrix}$$

Now,

$$AA^T = \begin{bmatrix} -4 & -7 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} -4 & 1 \\ -7 & 4 \end{bmatrix} = \begin{bmatrix} 65 & -32 \\ -32 & 17 \end{bmatrix}$$

Finding the eigenvector for AA^T .

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} (65 - \lambda) & -32 \\ -32 & (17 - \lambda) \end{vmatrix} = 0$$

$$\therefore (65 - \lambda) \times (17 - \lambda) - (-32) \times (-32) = 0$$

$$\therefore (1105 - 82\lambda + \lambda^2) - 1024 = 0$$

$$\therefore (\lambda^2 - 82\lambda + 81) = 0$$

$$\therefore (\lambda - 1)(\lambda - 81) = 0$$

$$\therefore (\lambda - 1) = 0 \text{ or } (\lambda - 81) = 0$$

$$v_1 = [-2, 1]$$

Eigenvectors for $\lambda = 1$ are:

$$v_2 = [0.5, 1]$$

For Eigenvector-1 (- 2, 1), Length $L = \sqrt{(-2)^2 + 1^2} = 2.236$

So, normalizing gives $u_1 = \left(\frac{-2}{2.236}, \frac{1}{2.236} \right) = (-0.894, 0.447)$

For Eigenvector-2 (0.5, 1), Length $L = \sqrt{0.5^2 + 1^2} = 1.118$

So, normalizing gives $u_2 = \left(\frac{0.5}{1.118}, \frac{1}{1.118} \right) = (0.447, 0.894)$

Similarly, we can find the eigenvectors for $A'A$ as:

Eigenvectors for $\lambda = 81$ are:

$v_1 = [0.51]$ Eigenvectors for $\lambda = 1$ are: $v_2 = [-21]$

For Eigenvector-1 (0.5, 1), Length $L = \sqrt{0.5^2 + 1^2} = 1.118$

So, normalizing gives $v_1 = \left(\frac{0.5}{1.118}, \frac{1}{1.118} \right) = (0.447, 0.894)$

For Eigenvector-2 (- 2, 1), Length $L = \sqrt{(-2)^2 + 1^2} = 2.236$

So, normalizing gives $v_2 = \left(\frac{-2}{2.236}, \frac{1}{2.236} \right) = (-0.894, 0.447)$

$$\therefore \Sigma = \begin{bmatrix} \sqrt{81} & 0 \\ 0 & \sqrt{1} \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore U = [u_1, u_2] = \begin{bmatrix} -0.894 & 0.447 \\ 0.447 & 0.894 \end{bmatrix}$$

V is found using formula $v_i = \frac{1}{\sigma_i} A^T \cdot u_i$

$$\therefore V = \begin{bmatrix} 0.447 & -0.894 \\ 0.894 & 0.447 \end{bmatrix}$$

U is found using formula $u_i = \frac{1}{\sigma_i} A \cdot v_i$

$$\therefore U = \begin{bmatrix} -0.894 & 0.447 \\ 0.447 & 0.894 \end{bmatrix}$$



Thank You